Logistic Regression for Titanic Survival Prediction

1. Introduction

Machine learning has a significant portion of applications when it comes to solving classification problems and yielding predictions based on historical data. Logistic regression is considered one of the most widely used supervised learning algorithms in binary classification tasks.

In this project, we developed a logistic regression model to predict if a passenger survived the Titanic tragedy according to specific passenger details such as age, sex, ticket cost, passenger class, family presence, and port of embarkation. The project consists of data analysis, preprocessing of data, training of the model, and analyzing its performance using various classification metrics.

2. Problem Statement

The purpose of this project is to create a Machine Learning model that can predict whether a passenger stayed alive or died in the catastrophe of the Titanic.
The mode should classify all passengers on board into one of the following two types:

- 0 – Passenger Did Not Survive
- 1 – Passenger Survived

The goal of the project is to conduct some analysis on the information about the passengers, learn what important factors influence their survival, carry out training of a Logistic Regression model and analyze how well the model performs by using standard metrics of classification.

3. Dataset Description

The dataset contains information about 1309 passengers and consists of 9 columns.

Dataset Features

Feature	**Description**
Passengerid	Unique identifier of each passenger
Age	Age of the passenger
Fare	Ticket fare paid
Sex	Gender of the passenger (Encoded)
sibsp	Number of siblings/spouses aboard
Parch	Number of parents/children aboard
Pclass	Passenger class
Embarked	Port of embarkation (Encoded)
Survived	Target variable (0 = No, 1 = Yes)

Dataset Summary

- Number of Records: 1309
- Number of Features: 9
- Missing Values: Only 2 in the Embarked column
- Duplicate Records: 0
- Dataset Type: Binary Classification

4. Exploratory Data Analysis (EDA)

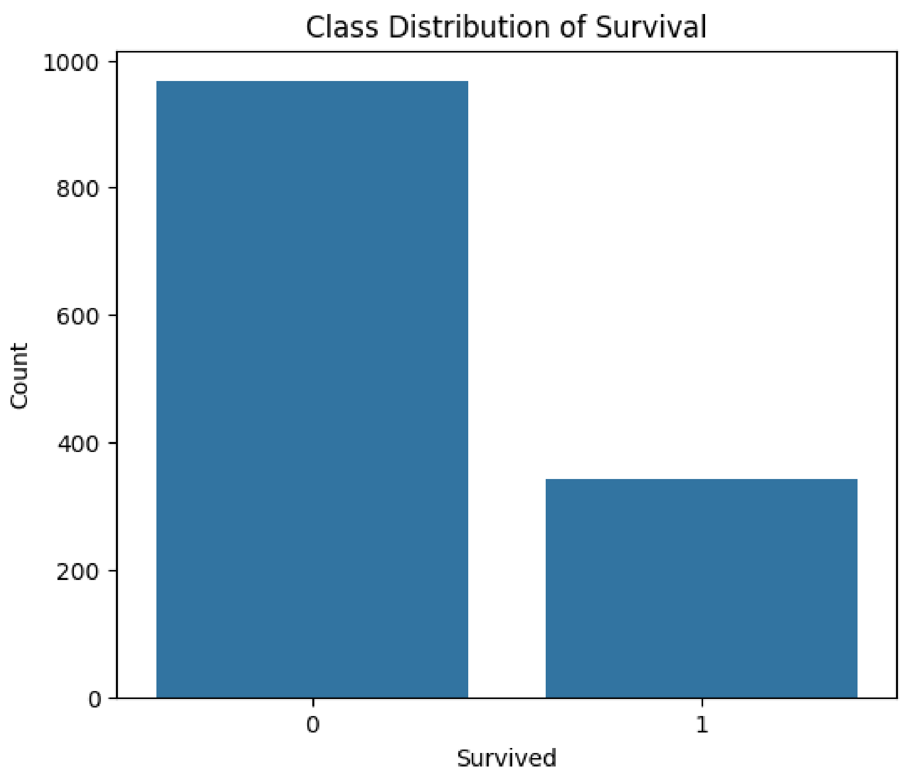
Exploratory Data Analysis was performed to understand the dataset before building the model.

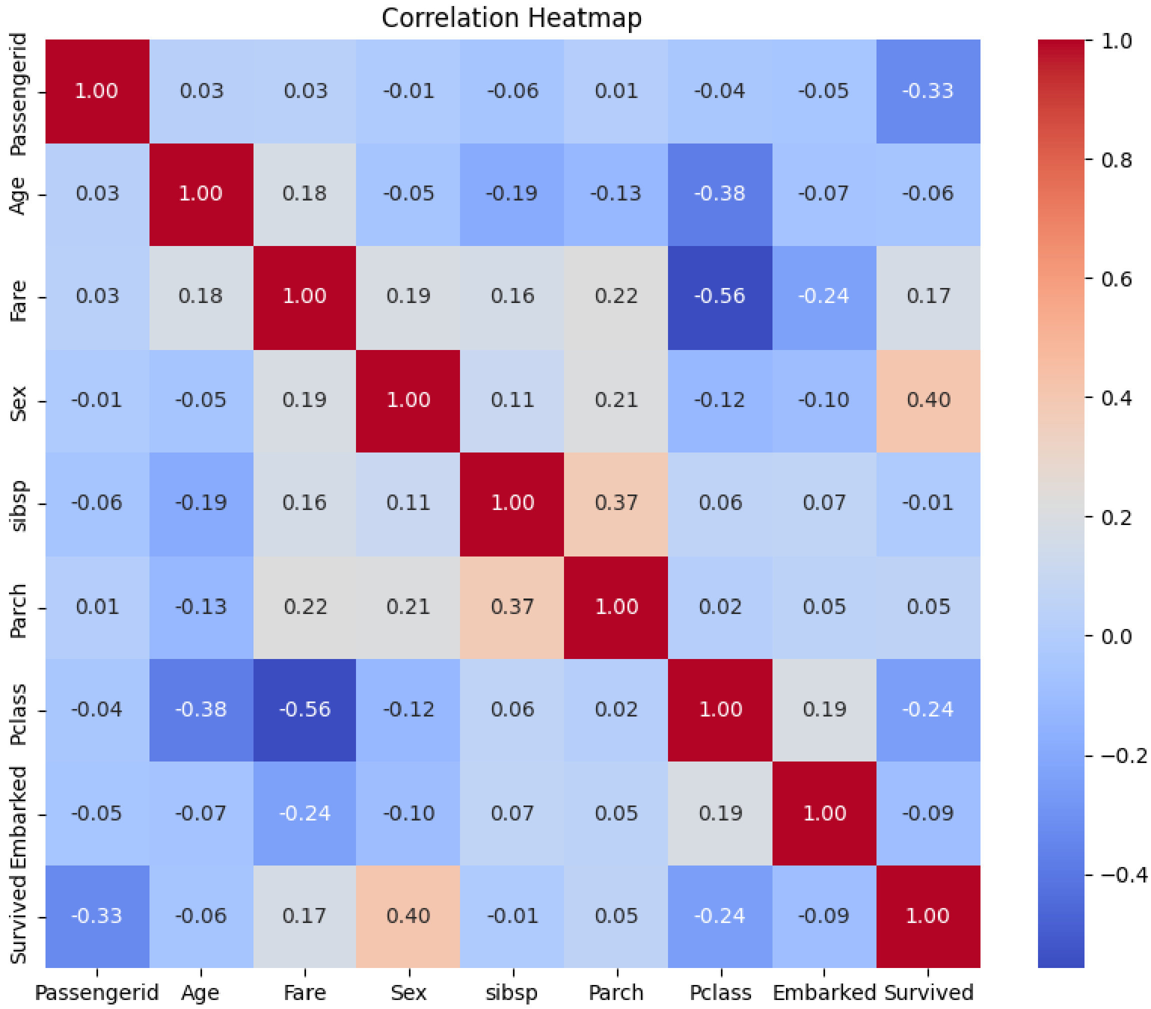
The following visualizations were created:

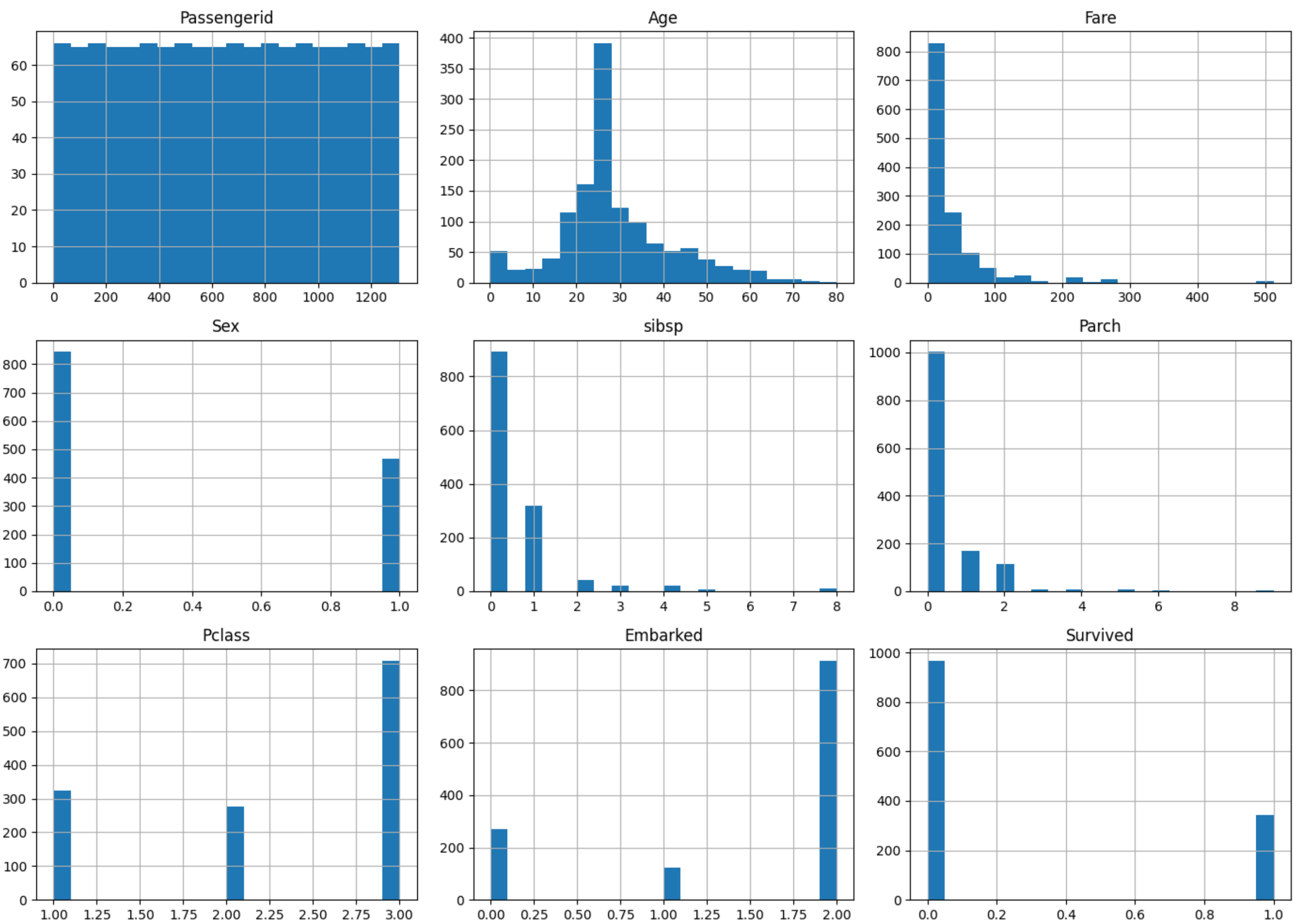
- Class Distribution
- Correlation Heatmap
- Histograms
- Boxplots
- Pairplot
- Outlier Analysis

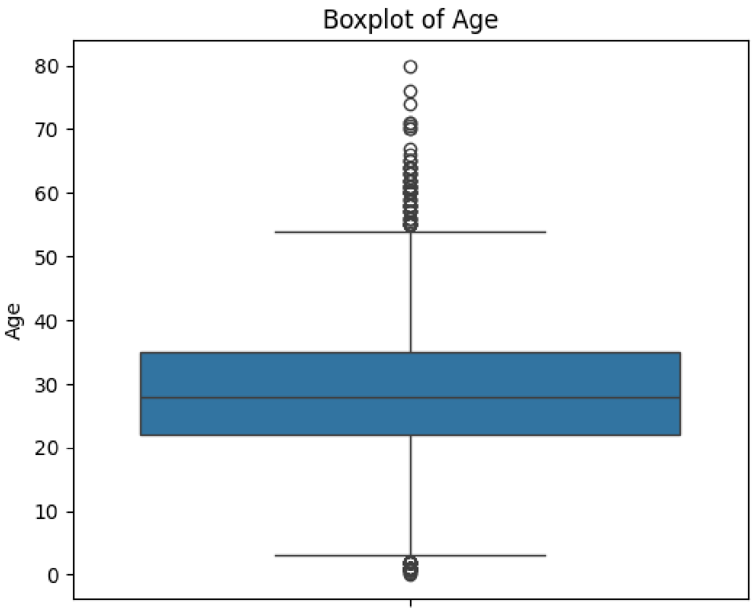
Observations

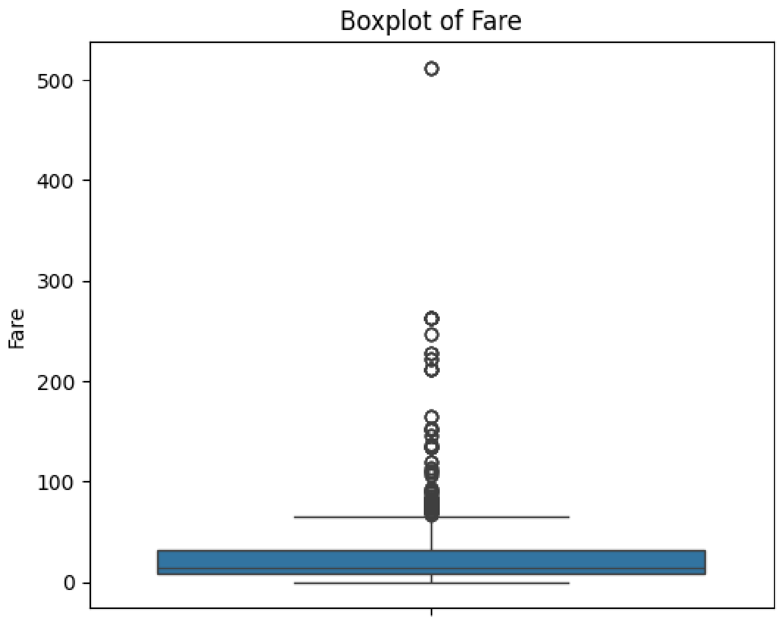
- The dataset contains an imbalance of the categories with more people dead than those alive.
- The sex of a passenger has the biggest positive correlation with the survival status of a passenger.
- Pclass of the passenger has a negative correlation with survival.
- Fare has a small positive correlation with survival.
- Age shows a little weak correlation with survival.
- The most important features revealed during EDA are Sex, Pclass, and Fare.

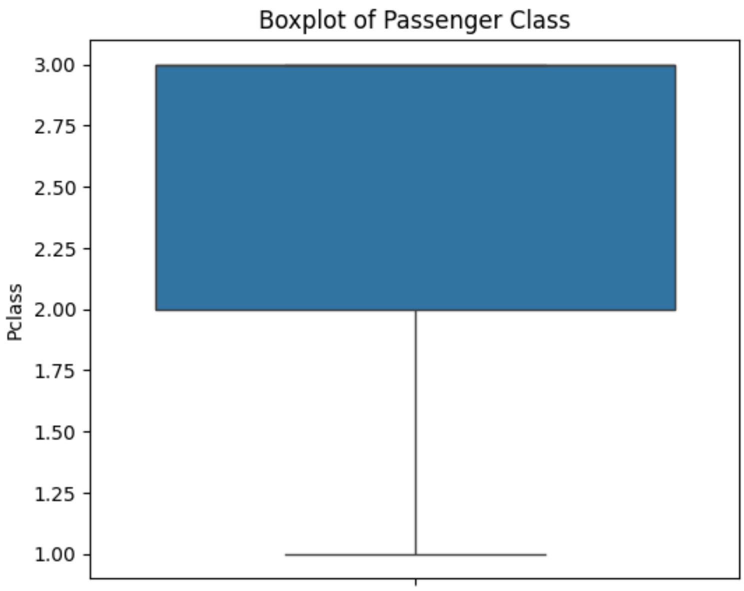


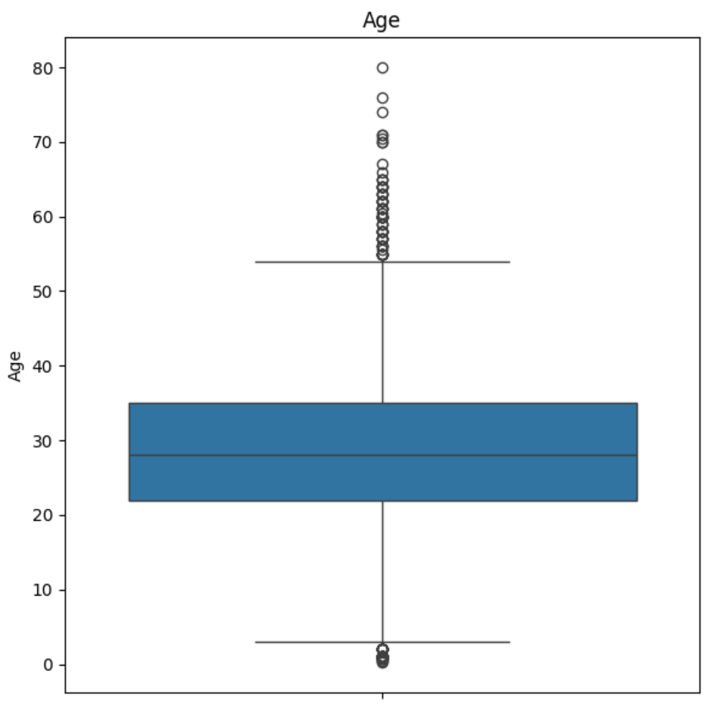

Correlation Heatmap
Passengerid 1.00 0.03 0.03 -0.01 -0.06 0.01 -0.04 -0.05 -0.33
Age 0.03 1.00 0.18 -0.05 -0.19 -0.13 -0.38 -0.07 -0.06
Fare 0.03 0.18 1.00 0.19 0.16 0.22 -0.56 -0.24 0.17
Sex -0.01 -0.05 0.19 1.00 0.11 0.21 -0.12 -0.10 0.40
sibsp -0.06 -0.19 0.16 0.11 1.00 0.37 0.06 0.07 -0.01
Parch 0.01 -0.13 0.22 0.21 0.37 1.00 0.02 0.05 0.05
Pclass -0.04 -0.38 -0.56 -0.12 0.06 0.02 1.00 0.19 -0.24
Embarked -0.05 -0.07 -0.24 -0.10 0.07 0.05 0.19 1.00 -0.09
Survived -0.33 -0.06 0.17 0.40 -0.01 0.05 -0.24 -0.09 1.00
Passengerid Age Fare Sex sibsp Parch Pclass Embarked Survived
1.0 0.8 0.6 0.4 0.2 0.0 −0.2 −0.4


Passengerid
Age
Fare
Sex
sibsp
Parch
Pclass
Embarked
Survived


Boxplot of Age
Age
0
10
20
30
40
50
60
70
80


Boxplot of Fare
Fare
0
100
200
300
400
500


Boxplot of Passenger Class
Pclass
1.00
1.25
1.50
1.75
2.00
2.25
2.50
2.75
3.00


Age
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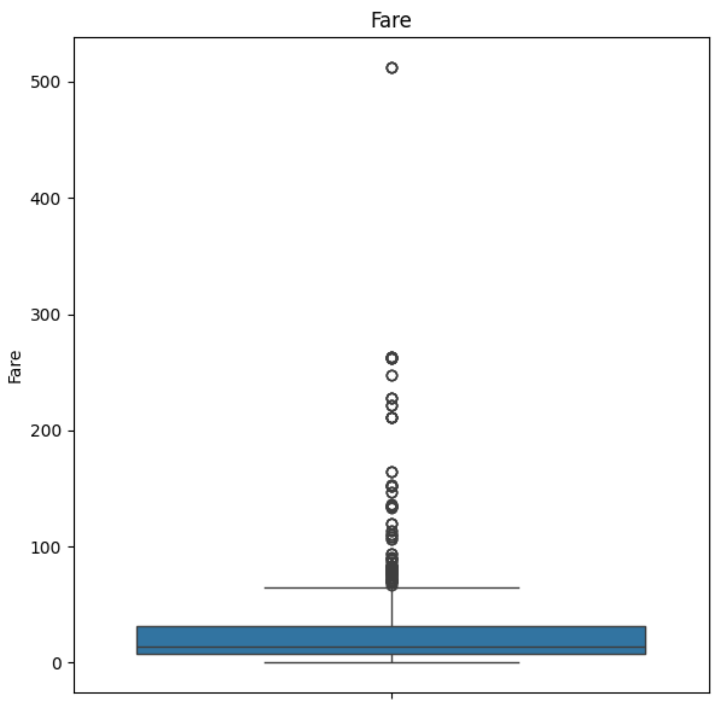

Fare
Fare
0
100
200
300
400
500

5. Data Preprocessing

Several preprocessing steps were performed before training the model.

Missing Value Handling

The missing values in the Embarked column were replaced using the most frequent value (Mode).

Label Encoding / One-Hot Encoding

No encoding was required because all categorical features were already encoded in the dataset.

Feature Selection

The Passengerid column was removed because it is only an identifier and does not contribute to prediction.

Feature Scaling

The features were standardized using StandardScaler so that all variables have similar scales, improving the performance of Logistic Regression.

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

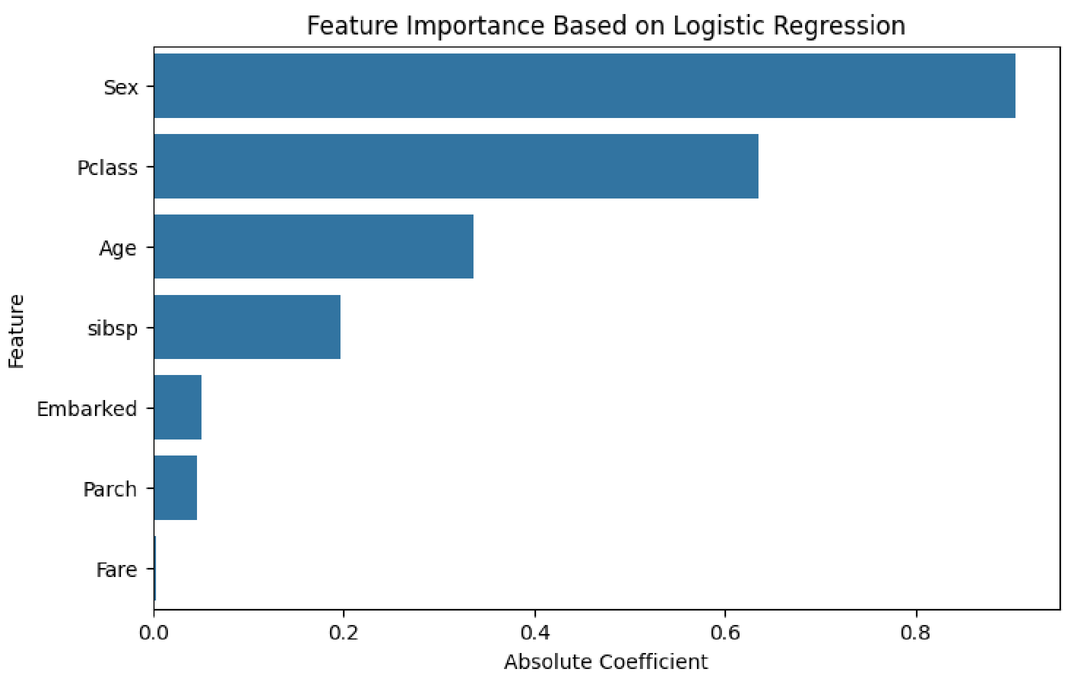
6. Model Building

A Logistic Regression model was developed using the Scikit-Learn library.The model was trained on the training dataset and then used to predict passenger survival on the testing dataset.

Based on the model coefficients, the most influential features are:

- Sex
- Pclass
- Age

The model intercept obtained was : **-1.3446**



7. Model Evaluation

The model was evaluated using different classification metrics.

Metric	**Value**
Accuracy	**76.72%**
Precision	**64.29%**
Recall	**36.99%**
F1 Score	**46.96%**

Feature Importance :

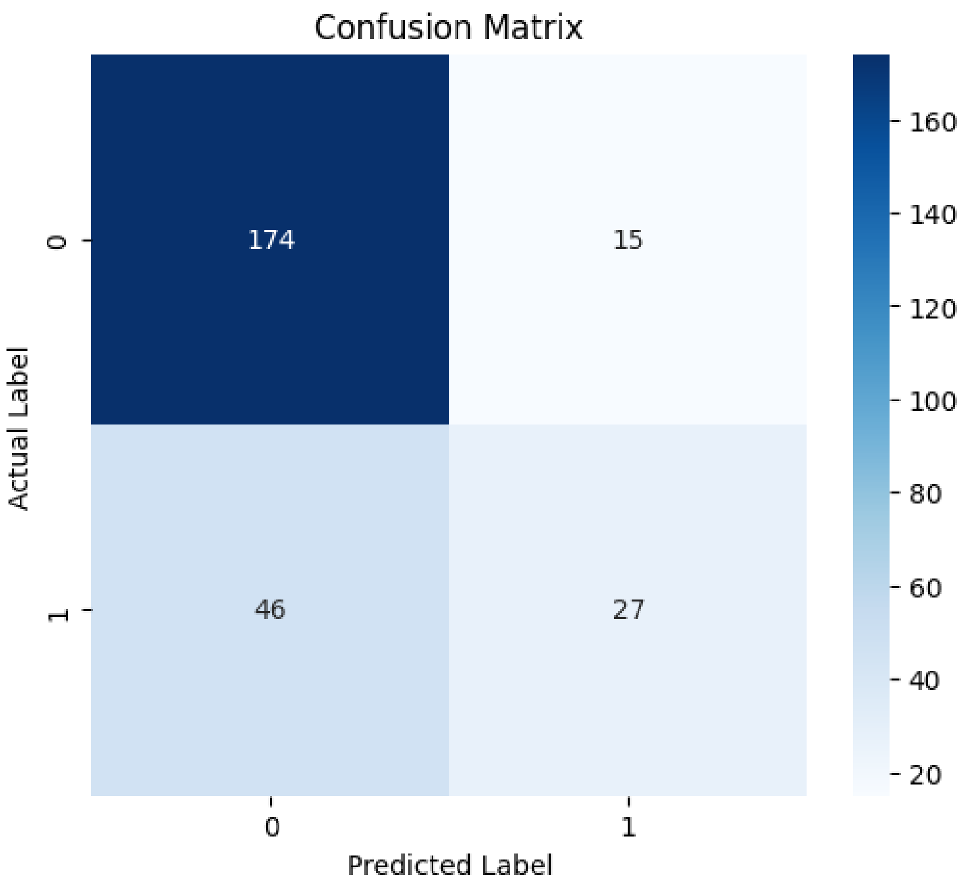
Rank	**Feature**	**Importance**
1	Sex	0.9053
2	Pclass	0.6349
3	Age	0.3365
4	sibsp	0.1960
5	Embarked	0.0512
6	Parch	0.0468
7	Fare	0.0038

Confusion Matrix :

Actual / Predicted	**0**	**1**
0	174	15
1	46	27

Observations

- The model achieved good overall accuracy.
- Precision is moderate, indicating that many predicted survivors were correctly classified.
- Recall is relatively low, meaning that several actual survivors were missed.
- The model performs better at predicting non-survivors than survivors because the dataset is imbalanced.



8. Hyperparameter Tuning

Hyperparameter tuning was accomplished using Logistic Regression.

The parameters below were applied in this work:

- Method: Logistic Regression
- Random State: 42
- Train-Test Split: 80 percent of data was used for training and 20 percent for testing.
- Feature Scaling: StandardScaler was utilized for feature scaling while building the model.
- Solver: Scikit-learn default solver was applied in this approach.

Parameters like type of solver, regularization parameter (C) and maximum number of iterations (max_iter), adjustments can be done to enhance model performance. Advanced techniques like Grid Search can be applied to identify appropriate hyperparameter values.

9. Business Insights

The following analysis has provided certain useful information:

- The gender of a passenger affects his/her chances of survival tremendously.
- Passengers traveling in higher luxury classes had better survival chances.
- The ticket price is also an important factor affecting survival chances.
- Age is not as important as gender and passenger class when it comes to predicting survival.
- This model indicates what factors affect passenger survival the most and can serve as an example of binary classification.
- Different classification models can be used in sectors like health care, insurance, banking, and transport for outcome prediction.
- Data received from this model will help organizations to see how important certain features are when making decisions based on data.

10. Conclusion

This project was successful in developing a Logistic Regression model for survival prediction of passengers on the Titanic.

The dataset was explored, cleaned, preprocessed, and employed for model training.

The model achieved an overall accuracy of 76.72% over the test dataset.

From the features analyzed, Sex, Pclass, and Age were found to be the most predictive for survival.

Despite the good performance of the developed model, the low recall indicates that some of the actual survivors were not identified. Further improvements may involve hyperparameter tuning, feature engineering, and applying more advanced classification methods, such as Random Forest, XGBoost, or SVM.